



CHANGE OVERVIEW FINAL VERSION 2.0 CONSOLIDATED

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INTRODUCTION

With effect from 5th July 2026 the eCoC (IVI) will become mandatory for the 2018/858 vehicle categories. It's important to note that the new IVI version and guideline related to the IVI 2.0 also apply to the 167/2013 and 168/2013 vehicle categories. Also, not all changes have been included in this document. This applies in particular to those amendments which have been the subject of discussion. In the appendix "Overview transition From IVI 1.10-To-2.0 v1.0" sheet more changes are indicated at field level.

*) Changes made after publication of the first draft IVI version 2.0. Other previously implemented changes are also included.

NAMESPACE

A namespace has been added to de XSD scheme.

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:sig="http://www.w3.org/2000/09/xmldsig#"
  xmlns="http://eu.ereg.initialvehicleinformation.v2" targetNamespace="http://eu.ereg.initialvehicleinformation.v2" elementFormDefault="qualified">
```

CHANGED DEFINITION FOR NON NUMERIC FIELDS

Some fields have been changed from numeric (NonnegativeInteger of PositiveInteger) to string field. As these fields are not used as numeric. This has no functional effect on the eCoC xml files.

<pre><xs:simpleType name="EnergySourceCode"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="2"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="EnergySourceCode"> <xs:restriction base="xs:string"> <xs:maxLength value="2"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="3"/> <xs:minInclusive value="0"/> <xs:maxInclusive value="279"/> </xs:restriction></pre>	<pre><xs:simpleType name="LoadCapacityIndex"> <xs:restriction base="xs:int"> <xs:minInclusive value="0"/> <xs:maxInclusive value="279"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:simpleType name="TyrePlyRatingNumber"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="2"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="TyrePlyRatingNumber"> <xs:restriction base="xs:string"> <xs:maxLength value="2"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:simpleType name="ColourCode"> <xs:restriction base="xs:positiveInteger"> <xs:totalDigits value="2"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="ColourCode"> <xs:restriction base="xs:string"> <xs:totalDigits value="2"/> </xs:restriction> </xs:simpleType></pre>

INTEGER DEFINITION

Due to remarks to be able to use generic tools to be able to use the XSD scheme in software generation tooling the integer definition has been changed. This has no functional effect on the eCoC xml files.

<pre><xs:simpleType name="Integer2"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="2"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="Integer2"> <xs:restriction base="xs:short"> <xs:minInclusive value="0"/> <xs:maxInclusive value="99"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:simpleType name="Integer3"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="3"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="Integer3"> <xs:restriction base="xs:short"> <xs:minInclusive value="0"/> <xs:maxInclusive value="999"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:simpleType name="Integer3"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="4"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="Integer4"> <xs:restriction base="xs:short"> <xs:minInclusive value="0"/> <xs:maxInclusive value="9999"/> </xs:restriction> </xs:simpleType></pre>

<pre><xs:simpleType name="Integer3"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="5"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="Integer5"> <xs:restriction base="xs:int"> <xs:minInclusive value="0"/> <xs:maxInclusive value="99999"/> </xs:restriction> </xs:simpleType></pre>
<pre><xs:simpleType name="Integer3"> <xs:restriction base="xs:nonNegativeInteger"> <xs:totalDigits value="6"/> </xs:restriction> </xs:simpleType></pre>	<pre><xs:simpleType name="Integer6"> <xs:restriction base="xs:int"> <xs:minInclusive value="0"/> <xs:maxInclusive value="999999"/> </xs:restriction> </xs:simpleType></pre>

CORRECTIONS

Value	Changed
Adressline	Addressline
MaximumPermissibleVerticalLoadCouplingPoint	Removed, to be stated as ValueS
InServiceMaximumPermissibleLadenMassAxleGroup	Renamed: InServiceMaximumPermissibleAxleGroupLadenMass
TypeVIIco2Weighted	Renamed: TypeVIIco2WeightedCombined
IviVersionNumberXsd	Value changed from 2.0.0 to 2.0
FuelingArrangementCode	Changed to English: FuellingArrangementCode
ExhaustEmissionLevel	Changed from EURO to Euro

GENERALCODEECOINNOVATION

Enumeration added: 28 / 29 / 32 / 33 / 35 / 37 / 38

List of eco-innovations approved under WLTP					
Passenger cars					
Eco-innovation code	Commission Implementing Decision (link)	Technology	Amendments	Powertrains (other than ICEV)	Fuels (other than diesel and gasoline)
28 ¹	CID (EU) 2019/1119	LED lights	CID (EU) 2020/1714	NOVC- HEV	Alternative fuels
37	CID (EU) 2020/174	Alternator	CID (EU) 2021/136	-	Alternative fuels
29	CID (EU) 2020/174	Alternator	CID (EU) 2021/488	-	Alternative fuels
32	CID (EU) 2020/1167	48V motor generator with DC/DC converter	CID (EU) 2021/488 CID (EU) 2022/252	NOVC- HEV	Alternative fuels
33	CID (EU) 2020/1232	12V motor generator	-	NOVC- HEV	Alternative fuels
38	CID (EU) 2022/716	Smart diesel fuel heater	-	NOVC-HEV	Only diesel
Light Commercial Vehicles					
Eco-innovation code	Commission Implementing Decision (link)	Technology	Amendments	Powertrains (other than ICEV)	Fuels (other than diesel and gasoline)
29	CID (EU) 2020/174	Alternator	CID (EU) 2021/488	-	Alternative fuels
32	CID (EU) 2020/1167	48V motor generator with DC/DC converter	CID (EU) 2021/488 CID (EU) 2022/252	NOVC- HEV	Alternative fuels
33	CID (EU) 2020/1232	12V motor generator	-	NOVC- HEV	Alternative fuels
35	CID (EU) 2020/1339	LED lights	CID (EU) 2020/1714 CID (EU) 2021/136	NOVC- HEV	Alternative fuels
38	CID (EU) 2022/716	Smart diesel fuel heater	-	NOVC-HEV	Only diesel

MECHANICALCOUPLINGCLASSCANBEFITTEDCODE

Changed enumeration

A			
A50-1	C50-7	J	S
A50-2	C50-X	K	T
A50-3	D	K1	W
A50-4	D50-A	K2	
A50-5	D50-B	K3	
A50-X	D50-C	K4	
B	D50-D	KA1	
B50-X	D50-X	KA2	
C	E	KA3	
C50-1	F	L	
C50-2	G	L1	
C50-3	G50	L2	
C50-4	G50-X	L3	
C50-5	H	L4	
C50-6	H50-X	L5	

VIENNACOUNTRYCODE

Enumeration restricted to official Vienna codes:

A	CR	GCA	LT	PL	SK	WL
AL	CV	GE	LV	PNG	SLO	WS
AND	CY	GH	M	PS	SME	WV
AM	CZ	GR	MA	PY	SN	YV
AUS	D	GUY	MAL	RA	SRB	Z
AZ	DK	H	MC	RC	SY	ZA
B	DOM	HK	MD	RCA	SYR	ZW
BD	DZ	HKJ	MEX	RCB	T	
BDS	E	HR	MGL	RCH	TG	
BG	EAK	I	MNE	RH	TJ	
BH	EAT	IL	MS	RI	TM	
BIH	EAU	IND	MW	RL	TN	
BR	EC	IR	MYA	RM	TR	
BRN	EG	IRL	N	RMM	TT	
BRU	EST	IRQ	NA	RN	UA	
BS	F	IS	NAM	RO	UAE	
BW	FIN	J	NIC	ROK	UK	
BY	FJI	JA	NL	ROU	USA	
C	FL	KH	NZ	RP	UZ	
CDN	FO	KG	OM	RSM	V	
CGO	GBA	KWT	P	RUS	VN	
CH	GBG	KZ	PA	RWA	WAG	
CI	GBJ	L	PK	S	WAL	
CL	GBM	LAO	PAL	SD	WAN	
CO	GBZ	LS	PE	SGP	WG	

ELECTRONICBRAKINGSYSTEM

Moved to the CoCDataGroup:

```
<xs:element name="ElectronicBrakingSystem" minOccurs="0" type="ElectronicBrakingSystemCode"/>
```

ENGINECAPACITYMIN/MAX

The fields EngineCapacityMin and EngineCapacityMax have no corresponding CoC field and have been removed from version 2.0

MECHANICALCOUPLINGTECHNICALLYPERMISSIBLETOWABLEMASS

According to 2015/504 the field has to be stated in Tonnes (T).

Change type from Integer 6 to Decimal 5.2.

DRIVINGDIRECTION

According to 2015/504 latest regulation this field is no longer part of the CoC. Has been removed from version 2.0.

MECHANICALCOUPLINGVALUEDC

Has been added after receiving comments from DAF.

PREVIOUS CHANGES DRAFT VERSION IVI 2.0

SEVERAL CHANGES IN ATTRIBUTE NAMING AND TYPE DEFINITIONS

Also see attached Excel overview "Overview transition From IVI 1.10-To-2.0 v1.1" for description of these changes and translation rules from IVI 1.x to IVI 2.0. Also the fields which are removed from version 2.0 and changes in field type, length and enumerations are indicated.

MANDATORY FIELDS IN IVI 2.0

These fields are mandatory for all vehicle categories .

- sig:Signature
As of version 2.0 the digital signature will be mandatory effective from 1st of September 2025.
The requirement of using an eIDAS advanced seal will be required effective 5th July 2026.
- IviReferenceld
- IviVersionNumberXsd
- IviVersionNumber
- IviVersionDateTime
- TypeApprovalCountry
- VehicleIdentificationNumber
- TypeApprovalType
- StageOfCompletion
- Type
- Variant
- Version if no Version is available in the Type Approval please state: N/A
this only applies to 167/2013 and 168/2013
- VehicleCategory
- TypeApprovalNumber
- TypeApprovalIssueDate
- SignerName
- SignerPosition
- SignatureLocation
- SignatureDate
- ManufacturerStageNumber
- ManufacturerName

Tables/Groups which have to be present (mandatory):

- CocDataGroup
- SigningAuthorityTable
- MakeTable
- ManufacturerTable
- GeneralConstructionGroup
- MassGroup
- DimensionGroup

Fields which are mandatory if the Group is applicable

These fields are mandatory if the Group is mentioned in the eCoC. The Group itself is not mandatory. For example in case of a EU Representative the Name is mandatory, if no EU Representative is mentioned in the Type approval the field is not mandatory and the entire Group does not have to exist in the eCoC.

Group	Fieldname
CommercialNameGroup	CommercialName
EURepresentativeGroup	EURepresentativeName
TowingBrakedTypeTrailGroup	MechanicalCouplingNumberTowingMasses
BodyworkTypeGroup	BodyworkTypeTrailer
GearRatioGroup	GearNumber
FinalDriveGroup	FinalDriveNumber
FinalDriveVehicleGroup	FinalDriveVehicleNumber
AxleGroup	AxleNumber
TyreAxleGroup	TyreNumber
TyreAxleMassCombinationGroup	TyreAxleCombinationNumber
AxleGroupGroup	AxleGroupNumber
AxleGroupMemberGroup	MemberAxleNumber
ColourGroup	Colour
CodeForBodyworkSupplementGroup	CodeForBodyworkSupplement
VehicleClassGroup	VehicleClass
MechanicalCouplingGroup	MechanicalCouplingNumber
EnergySourceGroup	EnergySource
TestFamilyIdentifiersGroup	TestFamilyIdentifier
	TestFamilyIdentifierValue
AdditionalEnergySourceGroup	AdditionalEnergySource
EcoInnovationsGroup	EcoInnovationCountry
	GeneralCodeEcoInnovation
TypeApprovalTransportDangerousGoodsGroup	TypeApprovalTransportDangerousGoodsClass
AdvancedSafetySystemsGroup	AdvancedSafetySystem
BallastSetGroup	BallastSetNumber
DriverSeatTypeCategoryGroup	DriverSeatTypeCategory
DriverSeatTypeCategoryClassGroup	DriverSeatTypeCategoryClass
TrackSetGroup	TrackSetNumber
TrackTrainSetRollerGroup	TrackTrainSetRollerNumber
ProvisionalApprovalTerritoryGroup	ProvisionalApprovalTerritoryCountry
RemarkGroup	RemarkSubjectDescription
	RemarkSubjectLanguage
	Remark

DESIGNATEDTYPEAPPROVALCOUNTRY

A new field has been added to the header and set to mandatory in order to be able to redirect eCoC's to the relevant Type approval authority easily. Without having to use the first part before the * of the TypeApprovalNumber. If this changes in future no change is needed in the software.

USAGE OF TAGNAMES "RANGE VALUES AND MINIMUM/MAXIMUM VALUES"

Added to Guideline document version 0.6

Example:

AxleSpacing
AxleSpacingAdjustableMin
AxleSpacingAdjustableMax

For the range values for masses Adjustable is not the correct naming, this has been changed to :
"Variable" -> example: MassInRunningOrderVariableMin

ENERGYSOURCE

Limit the number of EnergySources to the list as mention below:

10 Petrol
11 Petrol E5
12 Petrol E10
15 Ethanol
16 Ethanol E85
19 Mixture
20 Diesel
21 Biodiesel
22 ED95
30 LPG
40 CNG
44 Biomethane
50 Hydrogen
55 H2NG
60 LNG
90 Other
91 Compressed air
95 Electricity

Add 95 Electricity as an energy source. Keep the numbering of existing FuelCodes in line with the current numbering.

TYPE/VARIANT/VERSION

No pattern check will be added to limit the usage of other characters than A-Z and 0-9.

The field Type will be limited to 15 positions. No check on case sensitive check for Variant of Version.

TYPEAPPROVALISSUEDATE

Remove "RevisionDateVersion". The TypeApprovalIssueDate will be used to indicate the issue date of the document. This date doesn't change in case of revisions or corrections. It only changes in case of extensions. First set at extension 00 and subsequently at extensions 01 etc.

LOCATIONSTATUTORYPLATESCODE AND LOCATIONVIN

The fields will be replaced by a new Table.

LocationMarkingsTable

LocationMarkingsGroup

LocationMarkingsSubject

LocationMarkingsVehiclePart

LocationMarkingsVehiclePartSide

LocationMarkingsVehiclePartSideSection

LocationOfMarkingsSubjectCode

VIN = LocationVIN

STAT = LocationStatutoryPlatesCode

LocationOfMarkingsVehiclePartCode

ENGCT = Engine compartment

PASCT = Passenger compartment

LUGCT = Luggage compartment

INSPL = Instrument panel

DRAWB = Drawbar

CHASS = Chassis

APILR = A-pillar

BPILR = B-pillar

CPILR = C-pillar

DPILR = D-pillar

LocationOfMarkingsVehiclePartSideCode

FRONT = Frontside

LEFT = Leftside

RIGHT = Rightside

BACK = Backside

LocationOfMarkingsVehiclePartSideSectionCode

LEFT = Left

MIDDLE = Middle

RIGHT = Right

FRONT = Front

BACK = Back

LocationStatutoryPlates23Wheel and LocationVin23Wheel will not be added to this table in accordance with the definition in the 168/2013. The new table can be used for 2018/858 and 167/2013.

MANUFACTURER INFORMATION AND MULTISTAGE

Removed from the RoleTable and RoleGroup, multistage information has been simplified. This implies that ManufacturerStageNumber is a mandatory field also for non-multistage vehicles.

ManufacturerTable			
ManufacturerGroup			
ManufacturerStageNumber	1	2	3
ManufacturerName	First stage manufacturer name	Second stage manufacturer name	Third stage manufacturer name
ManufacturerAdressLine1	First stage address information	Second stage address information	Third stage address information
ManufacturerAdressLine2	First stage address information	Second stage address information	Third stage address information
ManufacturerAdressLine3	First stage address information	Second stage address information	Third stage address information
ManufacturerPlaceOfResidence	First stage address information place	Second stage address information place	Third stage address information place
ManufacturerCountryOfResidence	First stage address information country	Second stage address information country	Third stage address information country
PreviousStageTypeApprovalNumber		Type approval number previous stage	Type approval number previous stage
PreviousStageType		Type previous stage	Type previous stage
PreviousStageVariant		Variant previous stage	Variant previous stage
PreviousStageVersion		Version previous stage	Version previous stage

REPRESENTATIVE INFORMATION

Removed from the RoleTable and RoleGroup and stated in separate Group. Not added in a Table, as in all cases there's only one EU representative. If a representative is mentioned the EURepresentativeName is mandatory.

EURepresentativeGroup
 EURepresentativeName
 EURepresentativeAdressLine1
 EURepresentativeAdressLine2
 EURepresentativeAdressLine3
 EURepresentativePlaceOfResidence
 EURepresentativeCountryOfResidence

CONFIGURATION OF DOORS

Replace Text-40 field by a new definition.

ConfigurationOfDoorsGroup

NumberOfDoorsFront
 NumberOfDoorsLeft
 NumberOfDoorsRight
 NumberOfDoorsRear

Example:

	NumberOfDoorsFront	2
	NumberOfDoorsLeft	4
	NumberOfDoorsRight	3
	NumberOfDoorsRear	1

Please note: as the attribute “Number of doors” is still part of the CoC this will also be in the IVI file as NumberOfDoors in the GeneralConstructionGroup.

POSITIONOFSEATS

For the CoC data this element only applies to the 168/2013 vehicle categories. The proposal is not in line with the regulation. The proposed setup will be added to the regulation changes as proposed to the TCMV. Until the regulation has been changed no changes will be made for this field.

Regulation 901/2014

Bodywork

6.20.2.1. Door configuration and number of doors^(3g) (i) (j):

6.16.1. Number of seating positions:

6.16.1.1. Location and arrangement^(3g)(k):

(^k) Indicate the position by the following codes:

- rx: row number
- R: right side of the vehicle
- C: centre of the vehicle
- L: left side of the vehicle

Example for a vehicle with a first row with 2 front seating positions, 1 right, 1 left and a second row with 1 rear seating positions, 1 centre:

r1: 1R,1L r2: 1C

MASSDIMENSIONSGROUP

Has been divided into a MassGroup and a DimensionGroup.

```
<xsi:complexType name="MassGroup">
  <xsi:sequence>
    <xsi:element name="MassInRunningOrder" minOccurs="0" type="Integer6"/>
    <xsi:element name="MassInRunningOrderMinimum" minOccurs="0" type="Integer6"/>
    <xsi:element name="MassInRunningOrderMaximum" minOccurs="0" type="Integer6"/>
    <xsi:element name="ActualMass" minOccurs="0" type="Integer5"/>
    <xsi:element name="MassBaseVehicleInRunningOrder" minOccurs="0" type="Integer6"/>
    <xsi:element name="TechnicallyPermissibleMaximumLadenMass" minOccurs="0" type="Integer6"/>
    <xsi:element name="TechnicallyPermissibleMaximumLadenMassMinimum" minOccurs="0" type="Integer6"/>
    <xsi:element name="TechnicallyPermissibleMaximumLadenMassMaximum" minOccurs="0" type="Integer6"/>
    <xsi:element name="MinimumMassCompleted" minOccurs="0" type="Integer6"/>
    <xsi:element name="AdditionalMassAlternativePropulsion" minOccurs="0" type="Integer5"/>
    <xsi:element name="BallastMassTotal" minOccurs="0" type="Integer6"/>
    <xsi:element name="TechnicalPermissibleMaximumMassThreePointLift" minOccurs="0" type="Integer6"/>
    <xsi:element name="TechnicallyPermissibleMaximumStaticVerticalMassCouplingPoint" minOccurs="0" type="Integer6"/>
  </xsi:sequence>
</xsi:complexType>
<xsi:complexType name="DimensionGroup">
  <xsi:sequence>
    <xsi:element name="Wheelbase" minOccurs="0" type="Integer5"/>
    <xsi:element name="WheelbaseMinimum" minOccurs="0" type="Integer5"/>
    <xsi:element name="WheelbaseMaximum" minOccurs="0" type="Integer5"/>
    <xsi:element name="WheelbaseSideCar" minOccurs="0" type="Integer5"/>
    <xsi:element name="WheelbaseGroundRatio" minOccurs="0" type="Decimal32"/>
    <xsi:element name="Length" minOccurs="0" type="Integer5"/>
    <xsi:element name="LengthMinimum" minOccurs="0" type="Integer5"/>
    <xsi:element name="LengthMaximum" minOccurs="0" type="Integer5"/>
    <xsi:element name="Width" minOccurs="0" type="Integer4"/>
    <xsi:element name="WidthMinimum" minOccurs="0" type="Integer4"/>
    <xsi:element name="WidthMaximum" minOccurs="0" type="Integer4"/>
    <xsi:element name="Height" minOccurs="0" type="Integer4"/>
    <xsi:element name="HeightMinimum" minOccurs="0" type="Integer4"/>
    <xsi:element name="HeightMaximum" minOccurs="0" type="Integer4"/>
    <xsi:element name="SeatingHeight" minOccurs="0" type="Integer4"/>
    <xsi:element name="RearOverhang" minOccurs="0" type="Integer4"/>
    <xsi:element name="RearOverhangMinimum" minOccurs="0" type="Integer4"/>
    <xsi:element name="RearOverhangMaximum" minOccurs="0" type="Integer4"/>
    <xsi:element name="MaximumPermissibleRearOverhang" minOccurs="0" type="Integer4"/>
    <xsi:element name="MaximumPermissibleLength" minOccurs="0" type="Integer5"/>
    <xsi:element name="MaximumPermissibleLengthMinimum" minOccurs="0" type="Integer5"/>
    <xsi:element name="MaximumPermissibleLengthMaximum" minOccurs="0" type="Integer5"/>
    <xsi:element name="MaximumPermissibleWidth" minOccurs="0" type="Integer4"/>
    <xsi:element name="MaximumPermissibleWidthMinimum" minOccurs="0" type="Integer4"/>
    <xsi:element name="MaximumPermissibleWidthMaximum" minOccurs="0" type="Integer4"/>
    <xsi:element name="MaximumPermissibleHeight" minOccurs="0" type="Integer4"/>
    <xsi:element name="LengthOfTheLoadingArea" minOccurs="0" type="Integer5"/>
    <xsi:element name="LengthOfTheLoadingAreaMinimum" minOccurs="0" type="Integer5"/>
    <xsi:element name="LengthOfTheLoadingAreaMaximum" minOccurs="0" type="Integer5"/>
    <xsi:element name="ElongatedCabinIndicator" minOccurs="0" type="Indicator"/>
    <xsi:element name="AerodynamicDeviceEquipped" minOccurs="0" type="AerodynamicDeviceEquippedCode"/>
  </xsi:sequence>
</xsi:complexType>
```

RIMSIZEINCLUDINGOFFSET

RimSize

The size designation of wheels and rims shown by numbers and letters in following order as marked on the wheel:
Nominal rim diameter, Rim type (optional), Nominal rim width, Rim profile, Flange height (optional)
String-20

RimOffset

Offset, inset or outset as marked on the wheel, using the codes of ISO 3911:2021
Decimal value, can be a negative value.
Decimal 4.3 (-999.9)

WORKINGPRINCIPLECODE AND ENGINECYCLECODE

Change working principle codes to a combination of workingprinciple and enginecycle instead of repeating the engine cycle within the working principle.

WorkingPrincipleCode

Change enumeration to:

CA	Compressed air
ECI	External compression ignition
ECP	External positive ignition
ICI	Internal compression ignition
IPI	Internal positive ignition
TB	Turbine

WorkingPrincipleCode	WorkingPrinciple	CombustionCycle	FuelType
C2 compression ignition, two stroke	ICI	internal compression ignition	2ST
C4 compression ignition, four stroke	ICI	internal compression ignition	4ST
CA compressed air	CA	compressed air	
CI compression ignition	ICI	internal compression ignition	
DF dual fuel			
E2 positive ignition, two stroke	IPI	internal positive ignition	2ST
E4 positive ignition, four stroke	IPI	internal positive ignition	4ST
EC external combustion engine compression ignition	ECI	external compression ignition	
EE electric engine			
EF external combustion engine			
EP external combustion engine positive ignition	ECP	external positive ignition	
ER positive ignition, rotary	IPI	internal positive ignition	ROT
IC internal combustion engine compression ignition	ICI	internal compression ignition	
IE internal combustion engine positive ignition	IPI	internal positive ignition	
PI positive ignition	IPI	internal positive ignition	
TB turbine	TB	turbine	
CombustionCycle			
2ST = Two stroke			
4ST = Four stroke			
ROT = Rotary			
OTH = Other			

TYPEAPPROVALTRANSPORTDANGEROUSGOODSCLASS

Add enumeration for this field: TypeApprovalTransportDangerousGoodsClassCode

Text-10

EX/II
EX/III
AT
FL

MEMU

The enumeration value "OX" has been removed.

POWERTRAINGROUP

The powertrain level has been removed. Consolidated power fields have been moved to the CoC DataGroup. No vehicles with more powertrains exist.

LEVELS OF REGISTRATION

Registration of groups on different levels has to be avoided as the handling of the information and checking against the type approvals will make implementation to complex and almost impossible. Please see Appendix 1 for proposal how to handle the information for different vehicle types.

multiple levels. Field/Group IVI 1.x Field/Group IVI 2.0 Moved from Moved to TestFamilyIdentifiersGroup
TestFamilyIdentifiersGroup CoCDataGroup EnergySourceGroup MaximumSpeed MaximumSpeed FuelGroup
CoCDataGroup DeclaredMaximumSpeed DeclaredMaximumSpeedForward FuelGroup CoCDataGroup
DeclaredMaximumSpeedRearward DeclaredMaximumSpeedRearward FuelGroup CoCDataGroup
ExhaustEmissionLevelEuro ExhaustEmissionLevel FuelGroup CoCDataGroup NrBaseRegulActLastAmen
ExhaustEmissionLatestRegulation FuelGroup CoCDataGroup dMotVeh NrBaseRegulActLastAmen dEngines nNumber
Note: please see excel for more changes of levels of registration.

TESTFAMILYIDENTIFIERSTABLE

The TestFamilyIdentifiersTable has been moved to
CoCDataGroup/EnergyConvertorTable/EnergyConvertorGroup/EnergySourceTable
Mainly because ATCT Family can occur on an EnergySource level. In case of multiple fuels the FamilyIdentifiers
information has to be duplicated.

COMBINING FIELDS TEST RESULTS (N)OVC AND OTHER VEHICLE

Some fields are either for (N)OVC or other vehicles in the IVI message. For instance charge sustaining only for plug-in hybrid (OVC) or Co2combined only for NOVC vehicles. Field will not be combined to one tag. Main reason is that if these fields should be combined the definition is only clear when using other information in the IVI message. The tag itself does not define the content anymore. Having the fields separated is also in line with the legislation. If necessary the ICM can check the correct usage of the fields.

VECTOGROUP

Keep the Vecto related fields together in one VectoGroup:

CoCDataGroup

VectoGroup

- CryptographicHashManufacturerRecord
- ZeroEmissionHeavyDutyIndicator
- VocationalVehicleIndicator
- CryptographicHashCustomerInformationFile
- AveragePayload
- AverageOccupancy
- VehicleGroup
- VehicleSubgroup
- CryptographicHashVehicleFile
- CargoVolume

VolumeOrientationIndicator
EfficiencyRatioKilometer
EfficiencyRatioTonKilometer
EfficiencyRatioM3Kilometer

Please note that some Vecto fields are stated on an Energysource level and remain in that Group:

- SpecificCo2EmissionsMass
- SpecificCo2EmissionsPerson

PUREELECTRICVEHICLEINDICATOR

1. Question is if the field PureElectricVehicleIndicator according to point 23 is only applicable for hybrid vehicles that might or might not have a pure electric mode, see below for reference:

After re-evaluating the facts regarding the completion of field 23 in the CoC, I would like to inform you today about the changed procedure in relation to the type records and the IVI-CoC data.

Until now, the fields for the "pure electric drive" could also be filled with Yes (type data) or Y (IVI-CoC) in the case of **avehicle that is exclusively electrically driven** . According to a new perspective, this is not correct: "A pure electric vehicle does not have a selectable mode, it always drives purely electrically."

The points mentioned in the subject only refer to hybrid vehicles, so they are not to be used for electric vehicles!

- Use for **pure electric vehicles and combustion engines**:
Leave the field blank in the type data - in the IVI-CoC, the "PureElectricVehIndicator" element is omitted.
- Use for **hybrid vehicles** (where it is always to be used): Type data CoC Pt. 23 – Description sheet [Pt.3.4.3.1.1](#):
Type data:
Yes - switchable to purely electric
No - not switchable, can't drive purely electrically

IVI-CoC: „PureElectricVehIndicator“
Y - switchable to purely electric
N - not switchable, cannot drive purely electrically

2. Or does this field apply to fully electric vehicles only. So Yes only applies to vehicles with EnergySourceCode 95 for Electricity only.

Decision of the Forum is that the field has to be defined as the second option. If a switch mode (HybridElectricOperatingModeSwitchIndicator) is relevant information, a new field has to be added to the legislation and the IVI message.

IVI CONTENTCHECK MODULE (ICM)

The new version of the ICM will check if fields are relevant for a specific vehicle category. If not applicable the ICM will reject the IVI file.

For instance if a Colour is stated for the VehicleCategory M2, the file will be rejected.

Also if an indicator field is relevant for the VehicleCategory it has to be present with a Y or N.

For instance for M1 PureElectricVehicleIndicator is mentioned on the CoC, for Petrol vehicles the tag has to be present with value "N".

LANGUAGES IN IVI

Only the general remarks fields will be defined as multi lingual. Other free text fields will be defined without an language option. In general the CoC, according to the regulation, has to be published in one of the EU languages. Only for the fields mentioned below more than one language is optional:

- ✓ Remarks
- ✓ AdditionalInformation
- ✓ Exemptions
- ✓ Comments

CHANGED THE ENUMERATION FROM COUNTRY CODES TO LANGUAGE CODES.

Restricted to the official EU languages:

[Languages, multilingualism, language rules | European Union \(europa.eu\)](#) using the ISO-639-1 codes:

639-1	639-2	639-3	EU Languages
bg	bul	bul	Bulgarian
cs	cze/ces	ces	Czech
da	dan	dan	Danish
de	ger/deu	deu	German
el	gre/ell	ell	Greek
en	eng	eng	English
es	spa	spa	Spanish
et	est	est	Estonian
fi	fin	fin	Finnish
fr	fre/fra	fra	French
ga	gle	gle	Irish
hr	scr/hrv	hrv	Croatian
hu	hun	hun	Hungarian
it	ita	ita	Italian
lt	lit	lit	Lithuanian
lv	lav	lav	Latvian
mt	mlt	mlt	Maltese
nl	dut/nld	nld	Dutch
pl	pol	pol	Polish
pt	por	por	Portuguese
ro	rum/ron	ron	Romanian
sk	slo/slk	slk	Slovak
sl	slv	slv	Slovenian
sv	swe	swe	Swedish

SPEEDCATEGORYSYMBOLCODE

Extension of enumeration:

A1	D	N	Z
A2	E	P	
A3	F	Q	
A4	G	R	
A5	H	S	
A6	I	T	
A7	J	U	
A8	K	V	
B	L	W	
C	M	Y	

TYREFITTEDPRODUCTIONINDICATOR / TYRECO2DETERMINATIONINDICATOR

Relates to Item 35. This only states information on the fitted tyre/wheel combination which is used for Co2 determination. If TyreFittedProductionIndicator = Y this also indicates that it has been used for CO2Determination. The TyreCo2DeterminationIndicator has been removed.

EXHAUSTEMISSIONLEVEL AND EXHAUSTEMISSIONLATESTREGULATIONNUMBER

Have been moved to the CoCDataGroup. Valus not different on fuel/energysource level.

INTERCONNECTIONWITHPOWEREDAXLES

In the current version this is a text field. Suggestion is to change the free text to an standardized list of codes (enumeration).

Current version 1.x

InterconnectionWithPoweredAxleTable
InterconnectionWithPoweredAxleGroup
InterconnectionWithPoweredAxleNumber
InterconnectionPoweredAxes

New version 2.0

InterconnectionWithPoweredAxleTable
InterconnectionWithPoweredAxleGroup
InterconnectionWithPoweredAxleNumber
InterconnectionWithPoweredAxleConnection type ="InterconnectionWithPoweredAxleCode"

Possible values InterconnectionWithPoweredAxleCode:

TRA Transfer box
ELE Electronic
SHA Shaft

MECHANICALCOUPLINGCANBEFITTED FIELDS

The suggested field "MechanicalCouplingFittedIndicator" has been removed from draft IVI 2.0.

Item 44. Number of Approval Certificate or Approval Mark of coupling device (if fitted)

MechanicalCouplingApprovalNumber (if a value is stated here the coupling device has been mounted by the manufacturer)

Item 45. Types or classes of coupling devices which can be fitted

In use for incomplete vehicles

MechanicalCouplingTypeCanBeFitted
MechanicalCouplingClassCanBeFitted

TECHNICALLYPERMISSIBLEMAXIMUMSTATICVERTICALMASSCOUPLINGPOINT

In the previous draft IVI 2.0 version we had two fields concerning StaticVerticalMassCouplingPoint

Current IVI 2.0

Item 19. TechnicallyPermissibleMaximumStaticVerticalMassAtCouplingPoint

Only to be used if the coupling device has been mounted at the rear of the vehicle.

Item 19.1 TechnicallyPermissibleMaximumStaticVerticalMassOnCouplingPoint

Only to be used if the coupling device has been mounted at the front of the vehicle.

In the current version 2.0 this has been changed to:

Item 19.1 TechnicallyPermissibleMaximumStaticVerticalMassCouplingPoint

No difference in the two fields anymore as "AtCouplingPoint" only applicable for towing vehicles, "OnCouplingPoint" only for towed vehicles.

MAXIMUMAUTHORISEDDEVIATIONSINDICATOR (AKA : EXCEEDINGDIMENSIONSINDICATOR)

This field was removed from the previous version of IVI 2.0 but has been re-instated:

In accordance with the second subparagraph of Article 6(5) of Regulation (EU) 2018/858, an EU type-approval may be granted for vehicles exceeding the maximum authorised dimensions set out in point 1.1 of

Sections C, D and E of Part 2 of Annex XIII to this Regulation, in which case the remark 'maximum authorised dimensions derogation' shall be included in point 52 of the type-approval certificate and the certificate of conformity.

RECOMMENDED TYRE PRESSURE MIN/MAX

Fields are no longer part of the CoC data and have been removed. Only the L-category demands the tyre pressure and no min/max is mentioned.

TyreAxleTable TyreAxleGroup RecommendedTyrePressureMin
TyreAxleTable TyreAxleGroup RecommendedTyrePressureMax

NUMBER OF DOORS BACK

Suggested by ACEA. The tag has been renamed to NumberOfDoorsRear

VEHICLE SUBGROUP CODE

The enumeration has been extended according to the recent regulation.

TRAILER MASSES

Usage of the table structure for all vehicle categories, avoiding duplication of information.

The key-items are braking-system, bodywork type of the trailer and the maximum masses.

Benefits

- for all vehicle categories one way to handle this mass information and setup the eCoC;
- for all vehicle categories one way of using the information for the registration processes;
- can be easily extended by extending the braking type code without having to add new fields

Disadvantages

- more complex to initially setup for the M,N and O categories

The table below is to be used for M,N, O , without mentioning "BodyworkTypeTrailer", as this field is only relevant for agricultural vehicles.

The UND (Undefined) is not to be used here anymore.

Example:

Masses		
13. Mass in running order/13.2 Actual mass of the vehicle	: 2175 / 2184	kg
16. Technically permissible maximum masses		
16.1 Technically permissible maximum laden mass	: 2640	kg
16.2 Technically permissible mass on each axle: No.1/No.2/No.3	: 1380 / 1470 / NA	kg
16.4 Technically permissible maximum mass of the combination	: 3390	kg
18. Technically permissible maximum towable mass in case of		
18.1 Drawbar trailer	750 kg	
18.3 Centre-axle trailer	750 kg	
18.4 Unbraked trailer	750 kg	
19. Technically permissible maximum static vertical mass at the coupling point	: 75	kg

16.4 Technically permissible towable mass of the combination (3390 kg)

<u>TowingBrakedTypeTrailGroup</u>	1	Refers to the <u>MechanicalCouplingNumber</u>
<u>MechanicalCouplingNumberTowingMasses</u>	BRK	Braked
<u>BrakedTypeTrail</u>		
<u>BodyworkTypeTable</u>		
<u>BodyworkTypeGroup</u>		
<u>BodyworkTypeTrailer</u>	UND	Undefined
<u>TechnicallyPermissibleMaximumCombinationMass</u>	3390	
18.1 Drawbar trailer (750 kg) (braked)		
<u>TowingBrakedTypeTrailGroup</u>	1	Refers to the <u>MechanicalCouplingNumber</u>
<u>MechanicalCouplingNumberTowingMasses</u>	BRK	Braked: 18.1, 18.2 and 18.3 are always braked trailers
<u>BrakedTypeTrail</u>		
<u>BodyworkTypeTable</u>		
<u>BodyworkTypeGroup</u>	DB	Drawbar trailer
<u>BodyworkTypeTrailer</u>		
<u>TechnicallyPermissibleMaximumTowableMass</u>	750	
18.3 Centre-axle trailer (750 kg) (braked)		
<u>TowingBrakedTypeTrailGroup</u>	1	Refers to the <u>MechanicalCouplingNumber</u>
<u>MechanicalCouplingNumberTowingMasses</u>	BRK	Braked: 18.1, 18.2 and 18.3 are always braked trailers
<u>BrakedTypeTrail</u>		
<u>BodyworkTypeTable</u>		
<u>BodyworkTypeGroup</u>	DC	Centre-axle trailer
<u>BodyworkTypeTrailer</u>		
<u>TechnicallyPermissibleMaximumTowableMass</u>	750	
18.4 Unbraked trailer (750 kg) (unbraked)		
<u>TowingBrakedTypeTrailGroup</u>	1	Refers to the <u>MechanicalCouplingNumber</u>
<u>MechanicalCouplingNumberTowingMasses</u>	UNB	Unbraked
<u>BrakedTypeTrail</u>		
<u>BodyworkTypeTable</u>		
<u>BodyworkTypeGroup</u>		
<u>BodyworkTypeTrailer</u>	UND	Undefined
<u>TechnicallyPermissibleMaximumTowableMass</u>	750	

BodyworkTypeTrailer - BodyworkTypeTrailerCode:

UND has been removed from enumeration. For M,N,O and L this field is not to be declared anymore. For agricultural vehicles a value is always applicable. Also DF and DG have been removed, as these are not applicable here.

BrakedTypeTrail-BrakedTypeTrailCode :

UND has been removed from enumeration. For M,N,O and L this field is not to be declared anymore. For agricultural vehicles a value is always applicable. Also CSB , PNH, IPD have been removed, as these are not applicable here.